

Product Requirements Document: Restaurant Billing & Management Software

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1. Summary

A billing and restaurant management system built for one new restaurant. Staff take orders at the counter or on phones, orders go to the kitchen instantly, and bills are GST-correct and printed or sent on WhatsApp.

Customers can scan a QR code on the table to order and pay online, and anyone can view the menu from home.

Billing and kitchen orders must keep working when the internet is down. The system runs on a small server inside the restaurant, and a small cloud server handles everything a customer's phone touches.

2. Goals and non-goals

Goals

1. Every bill is fast (under 30 seconds for a normal table) and GST-correct.
2. No lost or wrong kitchen orders: every item reaches the right kitchen section.
3. Customers can order and pay from the table without waiting for a waiter.
4. The owner knows each day's sales by payment method, the best-selling dishes and GST due, without manual counting.
5. Billing never stops because of the internet, and no data is lost if a computer breaks.
6. The owner pays no yearly licence fee.

Non-goals (V1)

- Stock/inventory, recipes and food cost, purchases/suppliers (features 7–9): Later.
- Swiggy/Zomato integration (21), table booking (14), loyalty and offers engine (12, 15), delivery management (13): Later.
- Multiple branches (25).
- Native Play Store / App Store apps. Everything is a web app (installable PWA).
- Online takeaway/delivery ordering from the home menu (the home menu is view-only in V1).

3. Users and roles

Role	Where	What they do
Owner	Counter PC, own phone	Everything; menu, prices, staff, settings, reports, cancel/refund approval
Manager	Counter PC	Billing, discounts within a limit, cancellations with a reason, day close
Cashier	Counter PC	Take orders, print and settle bills, take payments
Waiter	Phone	Take table orders, send them to the kitchen, request the bill
Kitchen	Kitchen printer / screen	See orders by section, mark them ready
Customer (table)	Own phone, via table QR	View the menu, order, pay online or at the counter, call a waiter
Visitor (home)	Own phone, via link/QR	View the menu only

Staff log in with a 4–6 digit PIN. Each role sees only its own screens.

4. Releases

Release	Target	Contents
R0: Opening edition	Installed 7 Oct, trial service 8 Oct, opening on or before Navratni (about 11 Oct 2026)	The family will not use a ready-made POS, so our own software must run from day one. Only what a restaurant needs to open: MENU-1,2,3,5,6,7,8 · ORD-1,2,3,4,5 · KOT-1,2,4,5 · BILL-1,2,3,4,6,8,9 · STAFF-1 (PINs, roles) · REP-1,2,3 (basic) · SET-1..4 · SYS-1,3,5 · the static online menu (WEB-1,2,5). Payments at the counter: cash, card, and the free UPI QR (recorded as UPI). Waiter phone ordering (WAIT-1,2) only if it is ready by 6 Oct. Backup plan for opening week: a numbered paper bill book and KOT pads kept at the counter.
R1: Complete POS	about 3–4 weeks after opening	The rest of §5.1–5.7, 5.11, 5.12: add-ons, kitchen screen, split bill, activity log, cancellation reports, day close, expenses, print queue, cloud backup and restore test.
R2: Online	about 4–5 weeks after R1 (about December 2026)	QR table ordering, online payment, WhatsApp bills, home menu served from the live menu, owner phone view of today's sales. Needs the payment gateway and WhatsApp approvals, so apply in week 1.
Later	By request	Features marked "Later" by the family.

R0 day plan (from 29 Sep):

Dates	Work
29–30 Sep	Project setup, database, settings, menu screens. Family sends the menu, logo, GSTIN, FSSAI no., tables and kitchen sections. Hardware ordered.
1–3 Oct	Tables, orders, KOT printing to network printers.
4–5 Oct	Billing, GST, bill numbering, bill printing, payments.
6 Oct	Staff PINs, basic reports, automatic local backup, waiter phone layout.
7 Oct	Install at the restaurant: server, printers, Wi-Fi; load the real menu.
8 Oct	Staff training and a practice service with family as customers.
9–10 Oct	Fixes and buffer.
11 Oct	Open. I stay on site for the first days.

Payment milestones: ₹25,000 at start, ₹25,000 when the opening edition is live, ₹25,000 at R2 accepted. Support ₹2,000/month starts 3 months after go-live.

5. Functional requirements

Priority: **M** = must have for that release, **S** = should have, **C** = could have.

5.1 Menu (feature 3) — R1

ID	Requirement	P
MENU-1	Categories with display order (Starters, Main Course, Breads, Rice, Drinks, Desserts, etc.).	M
MENU-2	Item: name, category, price, veg / non-veg / egg, kitchen section, short description, optional photo.	M
MENU-3	Variants with their own price (Half / Full, Small / Large).	M
MENU-4	Add-ons / extras with price (extra cheese, extra butter).	S
MENU-5	Mark an item "not available" instantly. This applies everywhere, including the QR menu, within 5 seconds.	M
MENU-6	Item search by name and by short code (for example "PBM" for Paneer Butter Masala).	M
MENU-7	Price changes apply only to new orders; old bills keep their prices.	M
MENU-8	Tax setting per item (default 5%).	M

5.2 Tables and orders (features 1, 2) — R1

ID	Requirement	P
ORD-1	Table map/grid with statuses: free, occupied, bill printed, paid online. Shows running amount and time seated.	M
ORD-2	Order types: dine-in (table), takeaway, delivery (phone order, customer name + phone + address).	M
ORD-3	Add items by tapping; change quantity; per-item note ("less spicy").	M
ORD-4	Move an order to another table; merge two tables.	M
ORD-5	Remove a sent item only with a reason; manager PIN if the item was already sent to the kitchen.	M
ORD-6	Number of guests on the order (optional).	C

5.3 Kitchen orders / KOT (feature 4) — R1

ID	Requirement	P
KOT-1	"Send to kitchen" creates a KOT with only the new items, split by kitchen section (for example Main, Tandoor, Drinks).	M
KOT-2	Each section's KOT prints on its own printer: KOT no., table, waiter, time, items, notes.	M
KOT-3	Kitchen screen (optional per section): live list of tickets, oldest first, with "Ready" button and a sound on a new ticket.	S
KOT-4	Cancelled items print a clear "CANCEL" KOT.	M
KOT-5	Reprint a KOT.	M
KOT-6	QR orders are marked "QR" and "PAID" / "PAY AT COUNTER" on the ticket.	M (R2)

5.4 Billing and payments (feature 1) — R1

ID	Requirement	P
BILL-1	Bill shows: restaurant name, address, phone, GSTIN, FSSAI licence no., bill no., date/time, table/order type, cashier, items with qty/rate/amount, subtotal, discount, taxable value, CGST and SGST (rate and amount), round-off, grand total, SAC 996331 (to confirm with the CA), payment method.	M
BILL-2	Bill numbers are consecutive and unique within a financial year (for example 2026-27/000123), with no gaps. Numbering restarts every 1 April.	M
BILL-3	Bills are never deleted. A cancelled bill stays with status "cancelled", a reason and who cancelled it.	M
BILL-4	Discount by percent or amount, with a reason. Above a set limit, manager PIN is needed.	M
BILL-5	Service charge only if the owner turns it on, and it must be optional for the customer (not added automatically), as per consumer rules. Packing charge for takeaway.	S
BILL-6	Payment methods: cash (with change calculation), UPI, card, online (QR), and split across methods.	M
BILL-7	Split the bill by items or into equal parts.	S
BILL-8	Reprint any bill (marked "DUPLICATE").	M
BILL-9	Tax mode set in settings: regular GST (tax invoice, 5% = 2.5% CGST + 2.5% SGST) or composition (bill of supply, no tax).	M
BILL-10	Customer phone on the bill (optional), used for the WhatsApp bill.	M

5.5 Waiter phone app (feature 5) — R1

ID	Requirement	P
WAIT-1	Mobile layout: pick table → add items → send to kitchen.	M
WAIT-2	Works on staff's Android phones over restaurant Wi-Fi; installable to the home screen.	M
WAIT-3	Receives "customer is calling" alerts from the table QR.	S (R2)

5.6 Staff and security (feature 11) — R1

ID	Requirement	P
STAFF-1	Add/disable staff, set role and PIN.	M
STAFF-2	Activity log: who did what and when (cancellations, discounts, reprints, price changes, drawer opens). Only the owner can view it.	M
STAFF-3	Automatic log-out on shared devices after inactivity.	S

5.7 Reports and day close (features 10, 16, 17) — R1 (owner phone view R2)

ID	Requirement	P
REP-1	Today / date range: total sales, number of bills, average bill, by payment method (cash / UPI / card / online).	M
REP-2	Item-wise sales and top sellers; category-wise sales.	M
REP-3	GST report for the CA: taxable value, CGST, SGST per day and per month; export to Excel/CSV.	M

ID	Requirement	P
REP-4	Cancelled bills, discounts and removed items report.	M
REP-5	Day close: expected vs counted cash, difference noted.	S
REP-6	Expenses: record daily spending by category; monthly total.	S
REP-7	Owner's phone view of today's sales from anywhere (through the cloud).	S (R2)

5.8 QR table ordering and online payment (feature 19) — R2

ID	Requirement	P
QR-1	Each table has a printed QR code with a unique link including a signed table key (cannot be guessed or changed to another table).	M
QR-2	Customer sees the live menu (only available items), adds to cart, adds a note, places the order. No app or login needed.	M
QR-3	Orders are accepted only while staff have the table open (or first order on a free table needs staff approval — decide, see §10).	M
QR-4	Payment choice: pay online now, or pay at the counter.	M
QR-5	Online payment via the payment gateway (UPI, cards). An order is marked paid only after the server verifies the payment (signature and webhook), never on the browser's word.	M
QR-6	Paid QR orders go straight to the kitchen. Pay-at-counter orders go to the kitchen as well (configurable: or wait for staff approval).	M
QR-7	Staff see QR orders on the table with a "QR" and "Paid online" tag. Staff can add items; the bill shows the amount already paid and the balance.	M
QR-8	Customer can order again on the same table; all orders join the same table bill.	M
QR-9	"Call waiter" button.	S
QR-10	Refund (full or partial) from the POS by the owner, through the gateway.	M
QR-11	If the internet is down, the QR page shows "Please order with the waiter".	M
QR-12	Daily check: gateway settlements vs payments recorded.	S

5.9 Online menu for home (part of feature 20) — R0 (static), R2 (live)

ID	Requirement	P
WEB-1	Mobile menu page: restaurant name, address, timings, categories, items, prices, veg marks. View-only, no ordering.	M
WEB-2	"Call us" and "Directions" (Google Maps) buttons.	M
WEB-3	R0: static page edited by me. R2: served from the live menu, so price changes and "not available" show automatically.	M
WEB-4	Pages required by the payment gateway: About, Contact, Terms, Privacy, Refund & Cancellation.	M (before gateway KYC)
WEB-5	Shareable link and QR for Google Maps, Instagram, WhatsApp and posters.	M

5.10 WhatsApp bills (feature 6) — R2

ID	Requirement	P
WA-1	After a bill is paid, and the customer phone is entered, send the bill as a PDF on WhatsApp from the restaurant's number, using an approved utility template (WhatsApp Cloud API).	M
WA-2	QR customers get the bill automatically if they gave their number.	S
WA-3	Fallback when the API is unavailable: "Send on WhatsApp" button that opens WhatsApp with the bill link.	M
WA-4	Status shown on the bill (sent / delivered / failed) with a resend option.	S
WA-5	Customer consent is recorded when the phone number is taken.	M

5.11 Settings (feature 24) — R1

ID	Requirement	P
SET-1	Restaurant name, logo, address, phone, GSTIN, FSSAI no., bill footer text.	M
SET-2	Tax mode and rates, service charge (on/off), packing charge.	M
SET-3	Tables (number, names, areas).	M
SET-4	Kitchen sections and which printer each uses; bill printer.	M
SET-5	Discount limit without manager PIN.	S

5.12 Offline, backup and printing (features 22, 23) — R1

ID	Requirement	P
SYS-1	Billing, KOT, printing and reports work with no internet (local server on the restaurant Wi-Fi/LAN).	M
SYS-2	When the internet returns, data syncs to the cloud automatically; nothing is lost or duplicated.	M
SYS-3	Automatic backups: local copy every hour; encrypted cloud copy every night; keep 30 days.	M
SYS-4	Tested restore procedure: a new PC can be set up from the cloud backup in under 2 hours.	M
SYS-5	Network thermal printers (ESC/POS, 80 mm); cash drawer opens from the bill printer.	M
SYS-6	If a printer is offline, the ticket is kept in a queue with an alert and retried.	M

6. Non-functional requirements

Area	Requirement
Speed	Adding an item or opening a table responds in under 0.5 s on the LAN. A KOT prints within 3 s of "Send". QR order reaches the kitchen within 5 s.
Availability	The local system works 100% without internet. UPS keeps the server and bill printer running through short power cuts.
Devices	Counter: Windows PC or large tablet (Chrome/Edge). Waiters: Android phones (Chrome). Customers: any phone browser.
Language	English UI for V1; menu names can be in English and Hindi. Hindi UI: Later.

Area	Requirement
Security	HTTPS everywhere; staff PINs stored hashed; roles enforced on the server; signed table QR keys; payment verified server-side; secrets never in the browser; the local server is not open to the internet (it connects out to the cloud).
Data and privacy	Customer phone numbers are stored only with consent and used only for bills (DPDP Act). Data belongs to the restaurant.
Records	Bills and GST data kept at least 6 years (GST record-keeping rules; confirm with CA).
Usability	A new cashier can make a bill after 15 minutes of training. Large buttons, touch-friendly.

7. Architecture (summary)

- **Local server** (mini PC at the restaurant): .NET 8 API, MySQL, SignalR for live updates to the kitchen and waiters, print service for network printers. Staff apps (React + TypeScript PWA) are served from here over Wi-Fi.
- **Cloud server** (small VPS in India): .NET API + database for the QR ordering pages, home menu, payment gateway webhooks, WhatsApp sending, owner phone view and backups.
- **Sync:** the local server keeps an outbound, authenticated connection to the cloud. It pushes menu changes and table status up; the cloud pushes QR orders and payment confirmations down. No port forwarding at the restaurant.
- **Integrations:** payment gateway (Razorpay / Cashfree / PhonePe PG — choose in §10), WhatsApp Cloud API (Meta), Google Maps link.
- Details go in a separate technical design document.

8. Main data (outline)

Restaurant settings · Staff (role, PIN hash) · Category · Item · Variant · Add-on · Table · Order · Order line (item, variant, qty, price, note, KOT id, status) · KOT · Bill (number, FY, totals, tax, status) · Payment (method, amount, gateway ref, status) · Refund · Customer (phone, consent) · WhatsApp message · Expense · Activity log · Sync queue.

9. Success measures (first month after go-live)

- 0 lost KOTs and 0 wrong bill totals reported.
- Average bill settle time under 30 s.
- At least 20% of dine-in orders placed through the table QR.
- At least 50% of bills sent on WhatsApp.
- Day-close cash difference explained every day.
- 0 hours of billing stopped because of internet problems.

10. Open questions (need answers from the family / CA)

#	Question	Needed for
1	Restaurant name, logo, address, phone, opening hours.	R0
2	Full menu: categories, items, prices, Half/Full, veg/non-veg, photos (optional).	R0
3	Number of tables and their names/areas (AC hall, outdoor, etc.).	R1

#	Question	Needed for
4	GSTIN; regular GST or composition scheme? Correct SAC and rate — confirm with the CA.	R1
5	FSSAI licence number.	R1
6	Kitchen sections and number of printers (for example Main + Tandoor + Drinks).	R1
7	Service charge: yes or no? Packing charge for takeaway: how much?	R1
8	Discount limit a cashier can give without the owner.	R1
9	Their final choices on the feature sheet (Need Now / Later / Not Needed).	R1 scope
10	Which payment gateway? (compare current fees; KYC needs PAN, GST, current account, website policy pages).	R2
11	QR orders on a free table: allow straight away, or only after a waiter opens the table?	R2
12	Phone number to use for WhatsApp Business (must be verified with Meta).	R2
13	Domain name choice.	R0

11. Risks

Risk	Effect	Plan
Only about 12 days to build the opening edition	Bugs on opening day during the festival rush	Keep R0 scope strictly to \$4; freeze features on 6 Oct; practice service on 8 Oct; paper bill book and KOT pads as backup; developer on site in opening week.
Menu or hardware arrives late	Nothing to install or test	Menu and details needed by 30 Sep; printers by 3 Oct (test with one printer at home first).
Payment gateway or WhatsApp approval is slow	R2 delayed	Apply in week 1; policy pages ready early; launch QR with "pay at counter" first if needed.
Wrong GST setup	Tax trouble	CA reviews a sample bill and GST report before go-live.
Printer or network problems during rush hours	Service stops	Network printers, print queue with retry, UPS, 4G backup, spare paper.
Scope grows during build	Delays	Changes after R1 sign-off go to "Later" or are priced separately.
Single developer	Support gaps	Written setup and restore guide; remote access for support.

12. Acceptance for each release

- **R1:** a full practice service (at least 30 bills across dine-in, takeaway and delivery) with printed KOTs and bills; GST report matches the bills; internet unplugged for 1 hour with no problems; backup restored on a second PC.
- **R2:** 10 test QR orders (5 paid online in gateway test mode, 5 pay-at-counter), one refund, 10 WhatsApp bills delivered; QR page falls back correctly with internet off.

13. Key user flows

Dine-in with waiter (R0)

1. Waiter or cashier opens a free table and adds items (with Half/Full and notes).
2. "Send to kitchen": a KOT prints at each kitchen section; the table turns occupied.
3. More items later: only the new items print as a new KOT.
4. Customer asks for the bill: cashier prints it; the table shows "bill printed".
5. Customer pays by cash, card or the counter UPI QR; cashier records the method; the table becomes free.

Takeaway / phone order (R0)

1. Cashier creates a takeaway or delivery order (name and phone; address for delivery).
2. Items are sent to the kitchen; packing charge added if set.
3. Bill printed and paid at pickup or on delivery.

QR table order with online payment (R2)

1. Customer scans the table QR; the live menu opens in the phone browser.
2. Customer adds items and a note, chooses "Pay now" and pays by UPI or card.
3. The cloud server verifies the payment with the gateway, then sends the order to the restaurant server.
4. KOT prints marked "QR · PAID"; staff see the order on the table with "Paid online".
5. Anything added later by staff is shown as the balance; the table is closed once the balance is zero.
6. The bill is sent on WhatsApp if the customer gave their number.

Day close (R1)

1. Manager counts the cash and enters it; the system shows expected vs counted.
2. Day report printed or viewed: sales by method, top items, cancellations, discounts.

14. Screens

Screen	Used by	Release
PIN login	All staff	R0
Tables (grid with status)	Cashier, waiter	R0
Order / billing (menu + current order + totals)	Cashier, waiter	R0
Bill preview and payment	Cashier	R0
Kitchen screen	Kitchen	R1
Menu management	Owner, manager	R0
Reports	Owner, manager	R0 (basic), R1 (full)
Settings (restaurant, tax, tables, printers, staff)	Owner	R0
Activity log, day close, expenses	Owner, manager	R1
QR codes (print table and home-menu QR)	Owner	R0 (home menu), R2 (tables)
Customer QR menu, cart, payment, order status	Customer	R2
Home menu (view-only)	Visitor	R0 (static), R2 (live)

The clickable demo (<demo/restaurant-demo.html>) shows the intended look of these screens.

15. Hardware and accounts

Item	Needed by
Mini PC server (i5, 16 GB RAM, SSD) + UPS	Install day (7 Oct)
80 mm thermal bill printer with LAN port + cash drawer	One printer by 1–2 Oct for testing
One 80 mm LAN thermal printer per kitchen section	Install day
Counter screen (touch preferred)	Install day
Wi-Fi router covering the whole dining area; broadband + 4G backup	Install day
Waiter phones (staff's own Android phones are fine)	Install day
Kitchen tablet/TV (optional)	R1
Domain name	R0 (online menu)
Cloud server (India region)	R0 for the menu page (static hosting is enough), R2 for full use
Payment gateway account in the restaurant's name	Apply in week 1; needed for R2
WhatsApp Business (Meta) verification + phone number	Apply in week 1; needed for R2

16. Glossary

Term	Meaning
POS	Point of sale: the billing software at the counter
KOT	Kitchen order ticket: the slip that tells the kitchen what to cook
GSTIN	The restaurant's GST registration number
CGST / SGST	Central and state parts of GST, each half of the total rate
Composition scheme	A simpler GST scheme where the restaurant pays a fixed rate and does not charge GST on the bill
SAC	Service code printed on the tax invoice
FSSAI	Food safety licence; its number must be printed on bills
Payment gateway	The company that takes online payments (UPI/cards) and sends the money to the bank
Webhook	A message from the payment gateway to our server confirming a payment
PWA	A website that can be added to the phone's home screen and works like an app
LAN	The restaurant's own local network (Wi-Fi and cables)